

Chapter 1

Introduction

This report outlines a novel use of the [Orthogonal Frequency Division Multiplexing \(OFDM\)](#) modulation technique as the backbone of the waveform for a [Radio Detection and Ranging \(RADAR\)](#) system as an aid to object classification. Some research into OFDM and RADAR as a whole is presented, followed by the design of the transmission system and the waveform itself, and then validation of the system is outlined, finally, tests are performed to determine the realistic performance of the system as a whole at classifying objects. In this introduction chapter, some background information is presented, followed by a problem statement, objectives, system requirements, and scope and limitations.

1.1 Background

RADAR technology has unlocked the ability for humans to be able to see the world around us with a new perspective. This ability to be able to see objects and more specifically, to see information such as position and velocity about these objects has helped with the development of devices and services that are of paramount importance to society. All the current use cases of RADAR do however lack the ability to determine exactly what object is being seen by the overall RADAR system. That is where the use of the OFDM waveform in a RADAR system becomes useful.

1.2 Problem Statement

Current RADAR technology works very well when determining the distance and direction of an object, however, it does not cope well with specific object classification tasks. One of the reasons for this is that current RADAR utilize either a pulse, continuous wave, or chirp waveform, which does not allow for object detection through statistical analysis due to the simple nature of the waveform. This is a drawback of using RADAR systems, since if object classification is desired, then a combination of radar and some other technology such as cameras would have to be used which increases the cost and complexity of the system as a whole.

1.3 Objectives

The goal of this project is to build a RADAR system that can simultaneously act as a normal RADAR which can determine the distance and direction of an object, as well as classify the specific object that it is looking at when compared to other example objects. The aim is to implement an OFDM based waveform into a RADAR system that would give the flexibility to analyze the returned signal statistically and determine what exactly the object is. The goal is to make a non-realtime system, that can distinguish between three different objects.

1.4 System Requirements

Since this system is very experimental in nature, the requirements are quite broad as of writing this, at the beginning of the project **OBVIOUSLY NEED TO CHANGE THIS EVENTUALLY**. The system needs to perform the following tasks.

1. Be able to operate as a normal RADAR. This entails determining range, azimuth, and elevation.
2. Be able to classify between three separate reflectors/scatterers.

1.5 Scope and Limitations

The scope of this project has been set to not include design of RADAR hardware specifically. This is because design of RADAR hardware is a complex task that would probably encompass an entire masters project in and of itself. Therefore, the RADAR hardware portion of the project will be handled by a [Software Defined Radio \(SDR\)](#). This extrapolates the complex RADAR hardware design aspect away from the project and allows me to focus on the data collection and processing aspect of the project.

Bibliography

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Glossary

OFDM Orthogonal Frequency Division Multiplexing

RADAR Radio Detection and Ranging

SDR Software Defined Radio